

Chapter 97

Deep Pixel Nodes

This chapter details Deep Pixel nodes found in Fusion. Deep Pixel nodes are capable of handling AOVs (Arbitrary Output Variables) from 3D-rendered files.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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Ambient Occlusion [SSAO]



The Ambient Occlusion node

Ambient Occlusion Node Introduction

Ambient Occlusion (AO) is the lighting caused when a scene is surrounded by a uniform diffuse spherical light source. Think of the scene as being surrounded by a humongous sphere that uniformly emits light from its surface. AO captures the low frequency lighting. It does not capture sharp shadows or Diffuse or Specular lighting. So, AO is usually combined with Diffuse and Specular lighting to create a full lighting solution.

The Ambient Occlusion node generates global lighting effects in 3D-rendered scenes as a post effect. It quickly approximates computationally expensive ray-traced global illumination. Being a post effect, it exposes similar aliasing issues like the Shader, Texture, and Volume Fog nodes. Hence, artifacts may appear in certain situations.

Usage

The AO node rarely works out of the box, and requires some tweaking. The setup process involves adjusting the Kernel Radius and Number Of Samples to get the desired affect.

The Kernel Radius depends on the natural “scale” of the scene. Initially, there might appear to be no AO at all. In most cases, the Kernel Radius is too small or too big, and working values must be found.

Inputs

There are three inputs on the AO node. The standard effect mask is used to limit the AO effect. The Input and Camera connections are required. If either of these is not supplied, the node does not render an image on output.

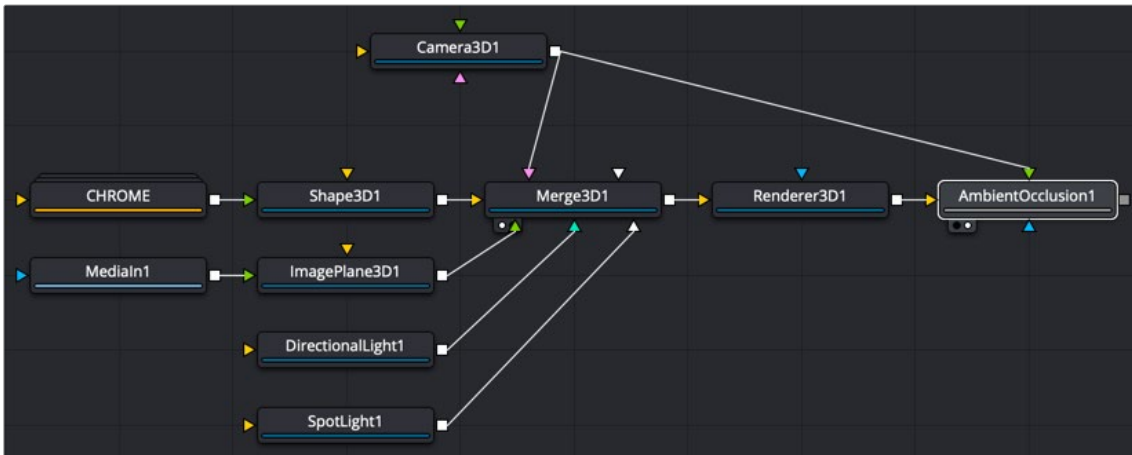
Input: This orange input accepts a 2D RGBA image, Z-Depth, and Normals.

Camera: The green camera input can take either a 3D Scene or a 3D Camera that rendered the 2D image.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Ambient Occlusion to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

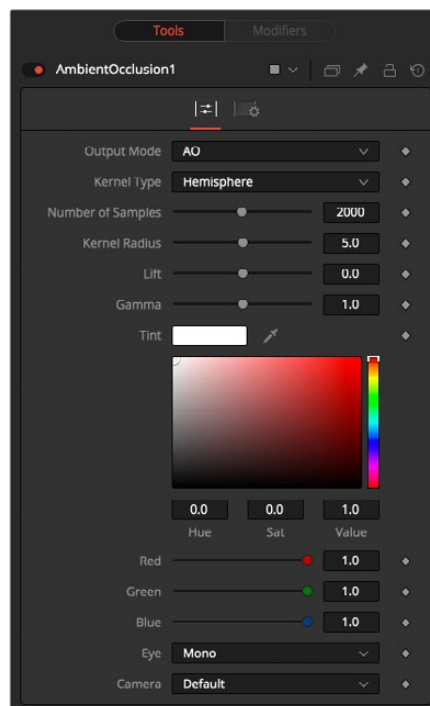
Basic Node Setup

The Ambient Occlusion node is typically placed after a Renderer 3D node. The Renderer 3D node must have Z-Depth and Normals enabled in its output channels. The Camera3D that is rendered by the Renderer3D node is then connected to the camera input on the AO node.



Ambient occlusion as a 2D post process

Inspector



Ambient Occlusion controls

Controls Tab

The controls tab includes all the main controls for compositing with AO. It controls the quality and appearance of the effect.

Output Mode

- **Color:** Using the Color menu option combines the incoming image with Ambient Occlusion applied.
- **AO:** This option outputs the pure Ambient Occlusion as a grayscale image. White corresponds to regions in the image that should be bright, while black correspond to regions that should be darker. This allows you to create a lighting equation by combining separate ambient/diffuse/specular passes. Having the AO as a separate buffer allows creative freedom to combine the passes in various ways.

Kernel Type

To determine the AO, rays are cast from a point on the surface being shaded, outward to a large enclosed sphere.

The AO factor is determined by the unoccluded rays that reach the sphere.

- **Hemisphere:** Rays are cast toward a hemisphere oriented to the surfaces normal. This option is more realistic than Sphere and should be used unless there is a good reason otherwise. Flat surfaces receive 100% ambient intensity, while other parts are darkened.
- **Sphere:** Rays are cast toward a sphere centered about the point being shaded. This option is provided to produce a stylistic effect. Flat surfaces receive 50% ambient intensity, while other parts are made darker or brighter.

Number of Samples

Increase the samples until artifacts in the AO pass disappear. Higher values can generate better results but also increase render time.

Kernel Radius

The Kernel Radius controls the size of the filter kernel in 3D space. For each pixel, it controls how far one searches in 3D space for occluders. The Filter Kernel should be adjusted manually for each individual scene.

If made too small, nearby occluders can be missed. If made too large, the quality of the AO decreases and the samples must be increased dramatically to get the quality back.

This value is dependent on the scene Z-depth. That means with huge Z values in the scene, the kernel size must be large as well. With tiny Z values, a small kernel size like 0.1 should be sufficient.

Lift/Gamma/Tint

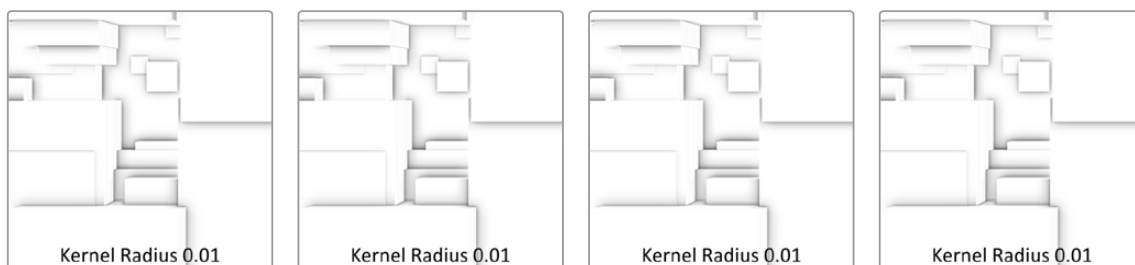
You can use the lift, gamma, and tint controls to adjust the AO for artistic effects.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Pixel nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

TIP: Combining multiple AO passes with different kernel radii can produce better effects.



AO Tips and Limitations

Transparency/Translucency: AO is designed to work with opaque objects. There are known limitations with transparent receivers and those with transparent occluders. You can work around some of these limitations by splitting out the transparent/translucent objects into separate scenes and only computing AO on the opaque objects.

Particles: Because of the transparency/translucency limitations, do not use AO on particles, unless the particles are solid opaque geometry. Anti-aliased edges are another form of transparency, so they also cause problems with AO.

Supersampling: To render anti-aliasing with Ambient Occlusion, enable HiQ for the Z and Normals pass in the Renderer 3D.

Viewer Dependence: AO methods work in viewer space, and the results are viewer dependent. This means the amount of darkening can vary depending on the view location, when in reality it should be constant. If at a point on an object the AO is 0.5, moving the camera could change it to 0.4.

Baking of AO: The OpenGL UV renderer can be used to bake AO into the textures on models.

Depth Blur [DBI]



The Depth Blur node

Depth Blur Node Introduction

The Depth Blur node is primarily used to create focal length or depth-of-field effects. It blurs 3D-rendered images based on included Z-channel values, and can also be used for general per-pixel blurring effects through the Blur Channel controls.

Inputs

The Depth Blur node includes three inputs: one for the main image, one for a blur image, and another for an effect mask to limit the area where the depth blur is applied.

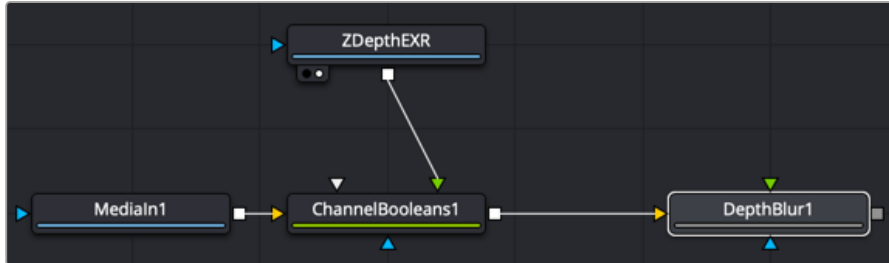
Input: This orange input is the only required connection. It accepts a 2D image that includes a Z channel. The Z channel is used to determine the blur amount in different regions of the image.

Blur Image: If the Blur Image input is connected, channels from the image are used to control the blur. This allows general 2D per-pixel blurring effects.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the depth blur to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

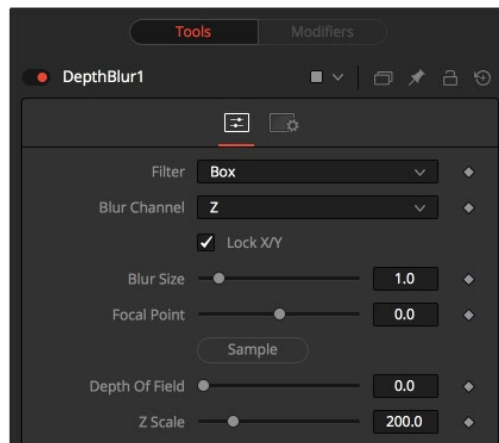
Basic Node Setup

The Depth Blur node receives the image containing the Z channel. Below, the Z depth channel is provided in a separate image file and combined with the RGB (beauty) image using a Channels Booleans tool. Channel Booleans sets the Z buffer channel to copy into the luminance foreground.



Depth Blur is applied to the beauty image based on a Z depth channel

Inspector



Depth Blur controls

Controls Tab

The Controls tab includes parameters for adjusting the amount of blur applied and the depth of the blurred area. It also includes options for selecting channels other than the Z channel for the blur map.

Filter

This menu selects the filter used for the blur.

- **Box:** This applies a basic depth-based box blur effect to the image.
- **Soften:** This applies a depth-based general softening filter effect.
- **Super Soften:** This applies a depth-based high-quality softening filter effect.

Blur Channel

Select one of these options to determine the channel used to control the level of blur applied to each pixel. The channel from the main image input is used, unless an image is connected to the node's green Blur Image input.

Lock X/Y

When toggled on, this control locks the X and Y Blur sliders together for symmetrical blurring.

Blur Size

This slider is used to set the strength of the horizontal and vertical blurring.

Focal Point

This control is visible only when the Blur channel menu is set to use the Z channel.

Use this control to select the distance of the simulated point of focus. Lowering the value causes the Focal Point to be closer to the camera; raising the value causes the Focal Point to be farther away.

Depth of Field

This control is used to determine the depth of the area in focus. The focal point is positioned in the middle of the region, and all pixels with a Z-value within the region stay in focus. For example, if the focal point were selected from the image and set to a value of 300, and the depth of field is set to 200, any pixel with a Z-value between 200 and 400 would remain in focus.

Z Scale

Scales the Z-buffer value by the selected amount. Raising the value causes the distances in the Z-channel to expand. Lowering the value causes them to contract. This is useful for exaggerating the depth effect. It can also be used to soften the boundaries of the blur. Some images with small depth values may require the Z-scale to be set quite low, below 1.0.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Pixel nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Fog [Fog]



The Fog node

Fog Node Introduction

The Fog node is used to create simulated fog effects on 3D-rendered images that contain a valid Z-buffer channel. The fog can be placed in front of or behind various elements of a rendered image based on the selected Z-channel planes.

Inputs

The Fog node includes three inputs: one for the main image with a Z channel, one for a blur image, and another for an effect mask to limit the area where the depth blur is applied.

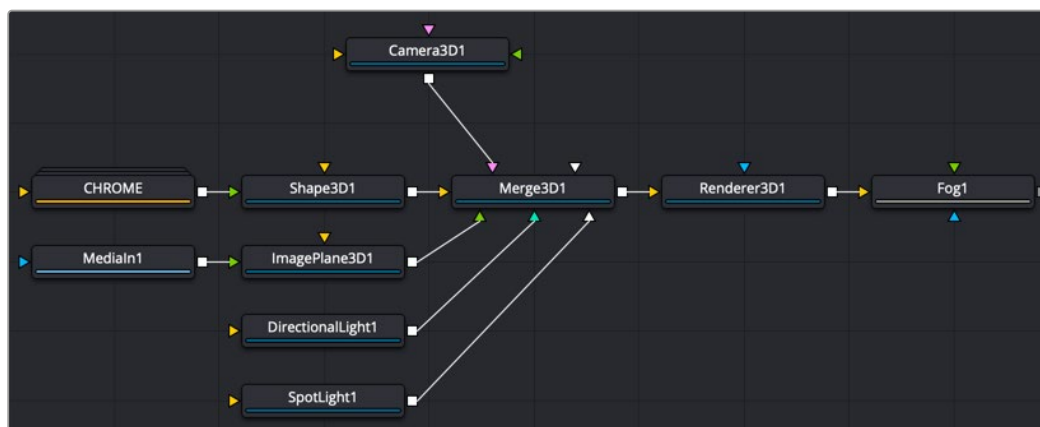
Input: This orange input is the only required connection. It accepts a 2D image that includes a Z channel. The Z channel is used to determine the fog amount in different regions of the image.

Blur Image: The green second image input connects an image that is used as the source of the fog. If no image is provided, the fog consists of a single color. Generally, a noise map of some sort is connected here.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the fog to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

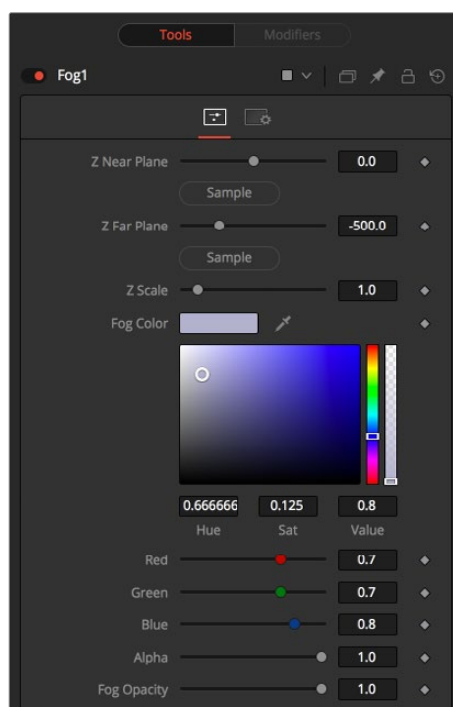
Basic Node Setup

The Fog node is typically placed after a Renderer 3D node. The Renderer 3D node must have Z-Depth enabled in its output channels.



A Fog node added as a 2D post process after the Renderer 3D node

Inspector



Fog controls

Controls Tab

The Controls tab includes parameters for adjusting the density and color of the fog.

Z-Buffer Near Plane and Far Plane

These controls are used to select the extents of the fog within the scene. To pick a value, drag the Pick button to an area on the image being viewed where the plane is to be located.

The Near Plane is used to select the depth where the fog thins out to nothing. The Far Plane is used to select the depth at which the fog becomes opaque.

Z Depth Scale

This option scales the Z-buffer values by the selected amount. Raising the value causes the distances in the Z-channel to expand, whereas lowering the value causes the distances to contract. This is useful for exaggerating the fog effect.

Fog Color

This option displays and controls the current fog color. Alpha adjusts the fog's transparency value.

Fog Opacity

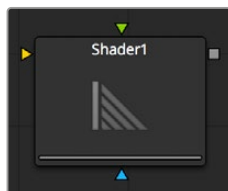
Use this control to adjust the opacity on all channels of the fog.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Pixel nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Shader [Shd]



The Shader node

Shader Node Introduction

The Shader node can control the lighting, reflection mapping, and 3D shading of elements in a rendered image. The node relies on the presence of the normal map channel in a rendered image. If this channel is not present, this node has no effect.

Inputs

The Shader node includes three inputs: one for the main image with normal map channels, one for a reflection map, and another for an effect mask to limit the area where the depth blur is applied.

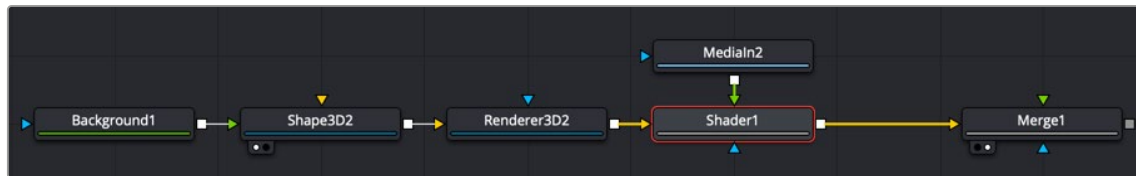
Input: This orange input is the only required connection. It accepts a 2D image that includes a normals channel.

Reflection Map Image: The green reflection map image input projects an image onto all elements in the scene or to elements selected by the Object and Material ID channels in the Common Controls. Reflection maps work best as 32-bit floating point, equirectangular formatted images

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the shader to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

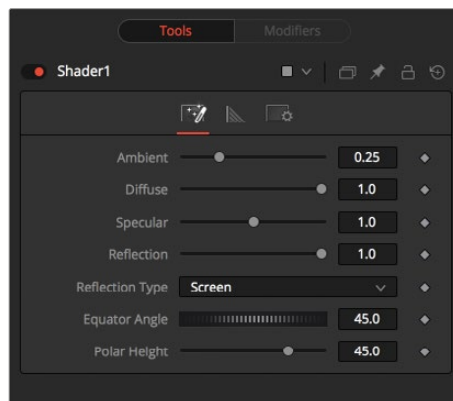
Basic Node Setup

The Shader node is inserted after a 2D image that contains a Normals channel. Below, a Renderer 3D is used to add a Normals channel to an image. The Shader node uses the normals for refining the surface appearance with a reflection map connected.



The Shader node using normals from a Renderer 3D and a reflection input

Inspector



Shader controls

Controls Tab

The Controls tab for the Shader node includes parameters for adjusting the overall surface reaction to light sources. You can modify the ambient, diffuse, specular, and reflection properties of the image connected to the orange image input.

Light Tab

The Controls tab includes parameters for basic lighting brightness and reflections.

Ambient

Ambient controls the Ambient color present in the scene or the selected object. This is a base level of light added to all pixels, even in completely shadowed areas.

Diffuse

This option controls the Diffuse color present in the scene or for the selected object. This is the normal color of the object, reflected equally in all directions.

Specular

This option controls the Specular color present in the scene or for the selected object. This is the color of the glossy highlights reflected toward the eye from a light source.

Reflection

This option controls the Reflection contribution in the scene or for the selected object. High levels make objects appear mirrored, while low levels overlay subtle reflections giving a polished effect. It has no effect if no reflection map is connected.

Reflection Type

This menu determines the type of reflection mapping used to project the image in the second input.

Screen: Screen causes the reflection map to appear as if it were projected on to a screen behind the point of view.

Spherical: Spherical causes the reflection map to appear as if it were projected on to a huge sphere around the whole scene.

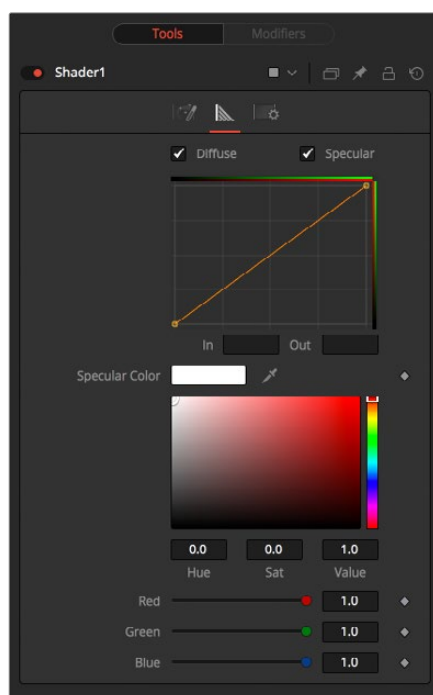
Refraction: Refraction causes the reflection map to appear as if it were refracting or distorting according to the geometry in the scene.

Equator Angle

Equator Angle controls the left to right angle of the light generated and mapped by the Shader node for the scene or the selected object.

Polar Height

Polar Height controls the top to bottom angle of the light generated and mapped by the Shader node for the scene or the selected object.



Shader tab controls

Shader Tab

The Shader tab is used to adjust the falloff of the Diffuse and Specular light and the tint color of the specular highlight.

Diffuse and Specular

When enabled, these checkboxes allow you to edit the Diffuse and/or Specular Shader curves in the Shader spline window.

In and Out

These options are used to display and edit point values on the spline.

Specular Color

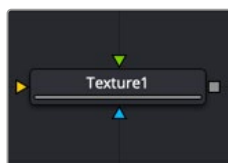
Use the Diffuse curve to manipulate the diffuse shading and the Specular curve to affect the specular shading. Drag a box over several points to group-select them. Right-clicking displays a menu with options for adjusting the spline curves.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Pixel nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Texture [Txr]



The Texture node

Texture Node Introduction

The Texture node controls the texture mapping of elements in a rendered image. The Texture node relies on the presence of U and V Map channels in a 3D-rendered image connected to the main Image input. If these channels are not present, this node has no effect.

NOTE: Background pixels may have U and V values of 0.0, which set those pixels to the color of the texture’s corner pixel. To restrict texturing to specific objects, use an effect mask based on the Alpha of the object or its Object or Material ID channel.

Inputs

The Texture node includes three inputs: one for the main image with UV map channels, one for a texture map image, and another for an effect mask to limit the area where the replace texture is applied.

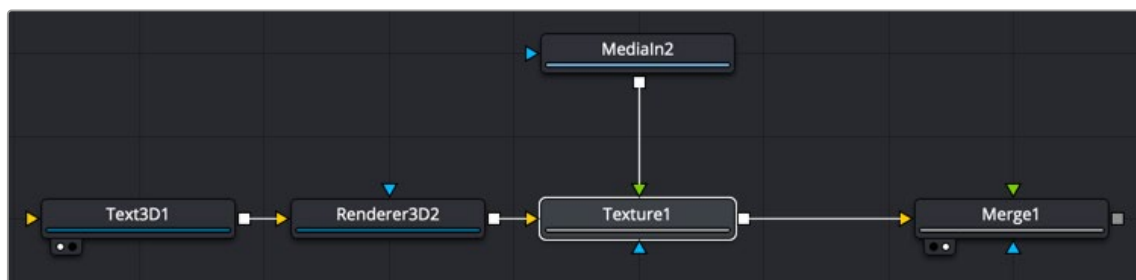
Input: This orange input accepts a 2D image that includes UV channels. If the UV channels are not in the images, this node has no effect.

Texture: The green texture map input provides the texture that is wrapped around objects, replacing the current texture.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the texture to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

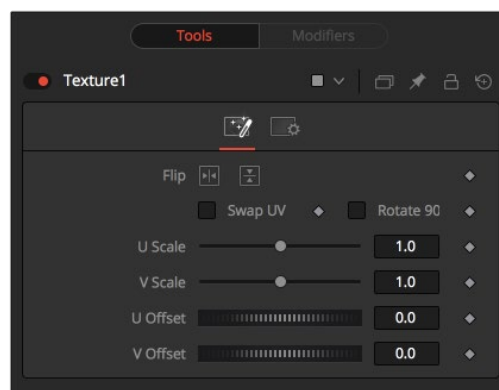
Basic Node Setup

The Texture node is inserted after a 2D image that contains a Texture UV channel. Below, a Renderer 3D is used to add texture coordinates to 3D text. The Texture node can then be used to manipulate those coordinates using a new texture connected to the green texture input.



A Texture node used to manipulate the texture coordinates and add texture to a Text 3D node

Inspector



Texture controls

Texture Tab

The Texture tab controls allow you to flip, swap, scale, and offset the UV texture image connected to the texture input.

Flip Horizontal and Vertical

Use these two buttons to flip the texture image horizontally and/or vertically.

Swap UV

When this checkbox is selected, the U and V channels of the source image are swapped.

Rotate 90

The texture map image is rotated 90 degrees when this checkbox is enabled.

U and V Scale

These controls change the scaling of the U and V coordinates used to map the texture. Changing these values effectively enlarges and shrinks the texture map as it is applied.

U and V Offset

Adjust these controls to offset the U and V coordinates. Changing the values causes the texture to appear to move along the geometry of the object.

Common Controls

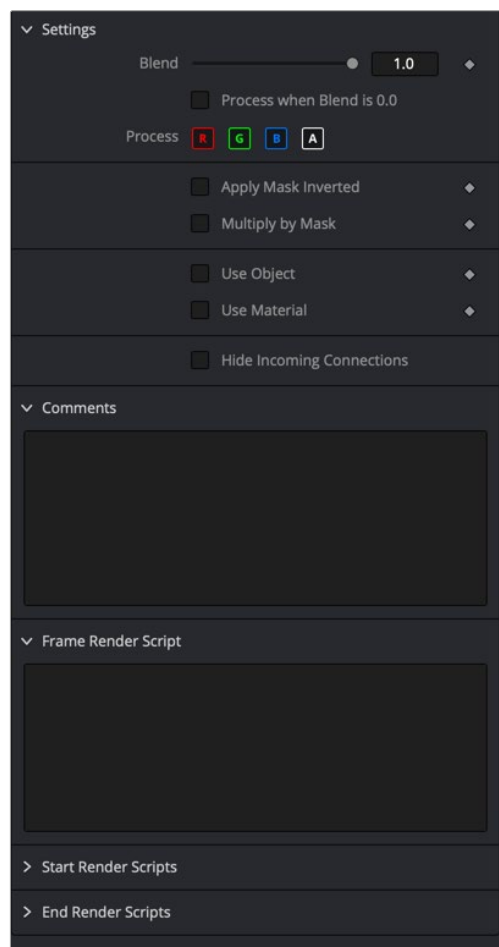
Settings Tab

The Settings tab in the Inspector is also duplicated in other Deep Pixel nodes. These common controls are described in the following “The Common Controls” section.

The Common Controls

Nodes that handle Deep Pixel compositing operations share several identical controls in the Inspector. This section describes controls that are common among Deep Pixel nodes.

Inspector



Common Settings tab in Deep Pixel Nodes

Settings Tab

The Settings tab in the Inspector can be found on every tool in the Deep Pixel category. The Settings controls are even found on third-party Deep Pixel-type plugin tools. The controls are consistent and work the same way for each tool although some tools do include one or two individual options that are also covered here.

Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task, but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a Blur tool is deselected, the blur is first applied to the image, and then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply by Mask

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not in the mask (i.e., set to 0) to become black/transparent.

Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels are used if present. The specific Material ID or Object ID affected is chosen using the next set of controls.

Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or *Chapter 18* in the Fusion Reference Manual.

Object ID/Material ID (Sliders)

Use these sliders to select which ID is used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off hardware-accelerated rendering using the graphics card in your computer. Enabled uses the hardware. Auto uses a capable GPU if one is available, but falls back to software rendering when a capable GPU is not available.

Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 causes Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Comments

The Comments field is used to add notes to a tool. Click in the field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.